



MATH1014 Calculus II Tutorial

Final Review

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1 A. Integration Applications

1.1 Fundamental Theorem of Calculus

1. If $F'(x) = f(x)$ on an interval, then

$$\int_a^b f(t) dt = F(b) - F(a).$$

2. If $A(x) = \int_a^x f(t) dt$ and f is continuous, then

$$A'(x) = f(x).$$

3. When you see an accumulated-area function, differentiate it. When you see a definite integral, look for an antiderivative.

1.2 Area Between Curves

1. Using vertical slices:

$$\text{Area} = \int_a^b (y_{\text{top}} - y_{\text{bottom}}) dx.$$

2. Using horizontal slices:

$$\text{Area} = \int_c^d (x_{\text{right}} - x_{\text{left}}) dy.$$

3. Always choose the variable that makes the region description simplest.
4. If top/bottom or right/left changes, split the interval at intersection points.

1.3 Volumes

1. Cross-sectional area method:

$$\text{Vol} = \int_a^b A(x) dx.$$

2. Disk/washer method: slices are perpendicular to the axis of rotation.

$$\text{Vol} = \int_a^b \pi (R(x)^2 - r(x)^2) dx$$

or the analogous dy version.



3. Cylindrical shells: slices are parallel to the axis of rotation.

$$\text{Vol} = \int 2\pi(\text{radius})(\text{height}).$$

4. Standard decision rule: first choose slice direction, then write radius and height from the geometry.

1.4 Arc Length

1. If $y = y(x)$ on $[a, b]$, then

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

2. If $x = x(y)$ on $[c, d]$, then

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy.$$

3. Before integrating, simplify the expression under the square root using identities or perfect-square structure.

1.5 Area of Surface Revolution

1. Rotating $y = f(x) \geq 0$ about the x -axis:

$$\text{Area} = 2\pi \int_a^b y ds, \quad ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

2. Rotating $x = g(y) \geq 0$ about the y -axis:

$$\text{Area} = 2\pi \int_c^d x ds, \quad ds = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy.$$

3. The safe workflow is: write ds , simplify, then choose a substitution if needed.

1.6 Work

1. Core template:

$$W = \int (\text{force}) \cdot (\text{distance}).$$

2. For fluids or lifting problems:

$$W = \int (\text{weight of slice}) \cdot (\text{lifting distance}).$$

3. For fluids, weight is usually $\rho g \cdot (\text{volume of slice})$.
4. A reliable setup is: choose a coordinate, describe one slice, find its volume/weight, then multiply by travel distance.



2 B. Integration Techniques

2.1 Integration by Parts

1. Formula:

$$\int u \, dv = uv - \int v \, du.$$

2. Choose u so that differentiating it makes it simpler.
3. Typical good choices for u include $\ln x$, inverse trig functions, and powers of $\ln x$.
4. If integration by parts reproduces the original integral, move that term to the left and solve for the integral algebraically.

2.2 Trigonometric Integrals

1. Use derivative relationships such as

$$\frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\cos x) = -\sin x.$$

2. Rewrite the integrand to create a useful du whenever possible.
3. Common identities:

$$\sin^2 x + \cos^2 x = 1, \quad \cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta.$$

4. Useful rewrites include

$$\sin x + \cos x = \sqrt{2} \sin\left(x + \frac{\pi}{4}\right), \quad 1 + \cos t = 2 \cos^2 \frac{t}{2}.$$

2.3 Trigonometric Substitution

1. For $\sqrt{1-x^2}$, use

$$x = \sin t, \quad dx = \cos t \, dt, \quad \sqrt{1-x^2} = \cos t.$$

2. For $\sqrt{1+x^2}$, use

$$x = \tan t, \quad dx = \sec^2 t \, dt, \quad \sqrt{1+x^2} = \sec t.$$

3. After substitution, simplify completely in trig form before integrating.
4. Convert the final answer back to the original variable.

2.4 Tangent Half-Angle Substitution

1. Let

$$u = \tan \frac{x}{2}.$$

2. Then

$$\sin x = \frac{2u}{1+u^2}, \quad \cos x = \frac{1-u^2}{1+u^2}, \quad dx = \frac{2}{1+u^2} du.$$

3. This substitution turns many rational expressions in $\sin x$ and $\cos x$ into rational functions of u .

2.5 Partial Fraction Decomposition

1. Factor the denominator first.
2. For linear factors, split into constants over each factor.
3. For irreducible quadratic factors, use a linear numerator.
4. Completing the square in a quadratic often leads to logarithm or arctan terms.

3 C. Extended Topics on Integration

3.1 Numerical Integration

1. Let $\Delta x = \frac{b-a}{n}$ and $x_i = a + i\Delta x$.

2. Left endpoint rule:

$$L_n = \Delta x [f(x_0) + f(x_1) + \cdots + f(x_{n-1})].$$

3. Right endpoint rule:

$$R_n = \Delta x [f(x_1) + f(x_2) + \cdots + f(x_n)].$$

4. Trapezoidal rule:

$$T_n = \frac{\Delta x}{2} [f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)].$$

5. Simpson's rule (requires even n):

$$S_n = \frac{\Delta x}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 4f(x_{n-1}) + f(x_n)].$$

3.2 Polar Coordinates

1. Conversion formulas:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r^2 = x^2 + y^2.$$

2. Area enclosed by a polar curve:

$$A = \int_{\alpha}^{\beta} \frac{1}{2} r^2 d\theta.$$

3. Arc length of a polar curve:

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta.$$

4. Always choose bounds that trace the curve exactly once, and use symmetry when it reduces work.

3.3 Improper Integrals

1. Type 1 improper integrals have infinite intervals:

$$\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx.$$

2. Type 2 improper integrals have an infinite discontinuity:

$$\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx$$

or the analogous one-sided limit.

3. Split the integral at any interior singular point.
4. The improper integral converges only if the corresponding limit exists and is finite.

3.4 Comparison for Improper Integrals

1. If $0 \leq f(x) \leq g(x)$ and $\int g$ converges, then $\int f$ also converges.
2. If $0 \leq f(x) \leq g(x)$ and $\int f$ diverges, then $\int g$ also diverges.
3. Reference p -integrals:

$$\int_1^\infty \frac{1}{x^p} dx \begin{cases} \text{converges,} & p > 1, \\ \text{diverges,} & p \leq 1, \end{cases}$$
$$\int_0^1 \frac{1}{x^p} dx \begin{cases} \text{converges,} & p < 1, \\ \text{diverges,} & p \geq 1. \end{cases}$$

4. A standard workflow is: identify where the problem occurs, compare with a simple p -type model, then conclude convergence or divergence.

4 D. Sequences and Series

4.1 Infinite Sequences

- A sequence converges if its terms approach a finite limit.
- **Squeeze Theorem:** if $a_n \leq b_n \leq c_n$ eventually and $a_n, c_n \rightarrow L$, then $b_n \rightarrow L$.
- **Monotone Convergence Theorem:** every bounded monotone sequence converges.
- For recursively defined sequences, first guess the limit, then prove monotonicity and boundedness.

4.2 Infinite Series and the Divergence Test

- A series $\sum a_n$ converges if its partial sums converge.
- Necessary condition:
$$\sum a_n \text{ convergent} \implies a_n \rightarrow 0.$$
- Therefore, if $\lim_{n \rightarrow \infty} a_n \neq 0$ or does not exist, then $\sum a_n$ diverges.
- If $a_n \rightarrow 0$, this test is inconclusive.

4.3 Telescoping Series

- Write the general term so cancellation becomes visible.
- Compute the partial sum S_n explicitly.
- Then evaluate $\lim_{n \rightarrow \infty} S_n$.

4.4 Integral Test

- If $a_n = f(n)$ where f is continuous, positive, and decreasing on $[1, \infty)$, then

$$\sum_{n=1}^{\infty} a_n$$

and

$$\int_1^{\infty} f(x) dx$$

either both converge or both diverge.

- This test is especially useful for logarithms and power-type expressions.

4.5 Comparison Tests for Series

- **Comparison Test:** for positive-term series, compare directly with a simpler benchmark.
- **Limit Comparison Test:** if

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c, \quad 0 < c < \infty,$$

then $\sum a_n$ and $\sum b_n$ have the same behavior.

- Common benchmark series:

$$\sum \frac{1}{n^p}, \quad \sum \frac{1}{ar^n}, \quad \sum \frac{1}{n(\ln n)^p}.$$

4.6 Absolute and Conditional Convergence

- A series $\sum a_n$ is absolutely convergent if $\sum |a_n|$ converges.
- Absolute convergence implies convergence.
- A series is conditionally convergent if $\sum a_n$ converges but $\sum |a_n|$ diverges.

4.7 Alternating Series Test

- For

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n, \quad b_n > 0,$$

the series converges if:

1. b_n is decreasing eventually;
 2. $b_n \rightarrow 0$.
- To show b_n is decreasing, it is often helpful to study a continuous function $f(x)$ with $f(n) = b_n$.



4.8 Ratio Test

- Let

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|.$$

- If $L < 1$, the series converges absolutely.
- If $L > 1$ or $L = \infty$, the series diverges.
- If $L = 1$, the test is inconclusive.
- This test is especially useful for factorials and exponentials.

4.9 Root Test

- Let

$$L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}.$$

- If $L < 1$, the series converges absolutely.
- If $L > 1$ or $L = \infty$, the series diverges.
- If $L = 1$, the test is inconclusive.
- This test is especially useful when the n th term already contains an n th power.

4.10 How to Choose a Series Test

- Use the divergence test first.
- For positive terms with logarithms or integrals that resemble functions of a real variable, try the integral test.
- For positive terms that look like a known benchmark, try comparison or limit comparison.
- For alternating signs, try the alternating series test and then check absolute convergence separately.
- For factorials or exponentials, try the ratio test.
- For n th powers, try the root test.

5 E. Power Series and Taylor Expansions

5.1 Power Series

- A power series centered at a has the form

$$\sum_{n=0}^{\infty} c_n (x - a)^n.$$

- There is a radius of convergence R such that the series converges for $|x - a| < R$.
- Inside its interval of convergence, a power series behaves like a polynomial.

5.2 Radius and Interval of Convergence

- Use the ratio test or root test to find R .
- After finding the open interval $|x - a| < R$, check both endpoints separately.
- Endpoint behavior may differ, so the final interval can be open, closed, or half-open.

5.3 Taylor and Maclaurin Series

- Taylor series centered at a :

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n.$$

- Maclaurin series is the special case $a = 0$.
- Important standard series:

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n, \quad |x| < 1,$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!},$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}, \quad -1 < x \leq 1.$$

5.4 Operations on Power Series

- Substitution: replace x by a simpler expression such as $2x$ or $-x^2$.
- Term-by-term differentiation:

$$\frac{d}{dx} \left[\sum c_n (x-a)^n \right] = \sum n c_n (x-a)^{n-1}.$$

- Term-by-term integration:

$$\int \left[\sum c_n (x-a)^n \right] dx = C + \sum c_n \frac{(x-a)^{n+1}}{n+1}.$$

- The radius of convergence stays the same after differentiation or integration, but endpoints must be checked again.

6 F. Vectors and 3D Geometry

6.1 Vectors: Magnitude and Unit Vectors

- For $\vec{v} = \langle v_1, v_2, v_3 \rangle$,

$$|\vec{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}.$$

- A unit vector in the same direction is

$$\frac{\vec{v}}{|\vec{v}|}.$$

6.2 Dot Product and Angle Between Vectors

- Algebraic definition:

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

- Geometric definition:

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta.$$

- Hence

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}.$$

6.3 Cross Product and Orthogonality

- The cross product $\vec{a} \times \vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{b}$.
- Compute it with the determinant formula:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

- It is a standard way to find a normal vector to a plane generated by two vectors.

6.4 Projection and Distance to a Plane

- Projection of $\vec{b}$ onto $\vec{a}$:

$$\text{Proj}_{\vec{a}} \vec{b} = \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2} \right) \vec{a}.$$

- If a plane through the origin is generated by $\vec{a}$ and $\vec{b}$, then a normal vector is

$$\vec{n} = \vec{a} \times \vec{b}.$$

- The distance from a point with position vector $\vec{r}$ to this plane is

$$d = \frac{|\vec{r} \cdot \vec{n}|}{|\vec{n}|}.$$